Introduction to Statistical Methods

(S2-25_AIMLCLZC418) – Assignment 1

AIML – Section 3

Question 1&2 carries 2.5 marks each.
Question 3 carries 5 marks
Total - 10 Marks
Duration: 30th May 2026 – 14th June 2026
1) Submissions are individual
2) Solve these on paper, scan, and upload
3) Plagiarism results in zero marks
4) Write your name, BITS ID and Section on each page
5) Only handwritten solutions with formula, full steps with proper justification are required.

Answer all the questions

1. Temperatures (in °C) measured at noon for 11 consecutive days are:
31, 29, 27, 34, 32, 28, 33, 29, 35, 26, 30.
(i) Compute the mean, median, variance, range, and Q1, Q3 for this group.
(ii) Use the IQR method to identify the outliers in the data set. **[2.5 M]**

2. In a class of 40 students, there are 18 boys and 22 girls. Out of the boys, 7 participate in sports and 6 scored an 'A' grade, with 3 boys involved in both. Among the girls, 9 participate in sports, 8 scored an 'A' grade, and 4 girls did both. If a student is picked at random, what is the probability the student is either involved in sports or scored an 'A' grade? **[2.5 M]**

3. A mental health organization is developing a simple machine learning model to help counsellors identify whether a person may have depression based on common symptoms. The organization decides to use the Naïve Bayes Classifier for prediction.

Person	Trouble Sleeping	Low Energy	Anxiety	Has Depression
A1	Yes	Yes	Yes	Yes
A2	No	Yes	No	No
A3	Yes	No	Yes	Yes
A4	No	No	No	No

A new patient visits the counsellor with the following symptoms:
• Trouble Sleeping = Yes
• Low Energy = No
• Anxiety = Yes

(a) Identify the target variable (class label) used in the dataset.

(b) Calculate the prior probabilities of:
1. Has Depression = Yes
2. Has Depression = No

(c) Find the following conditional probabilities:
1. P (Trouble Sleeping = Yes | Depression = Yes)
2. P (Low Energy = No | Depression = Yes)
3. P (Anxiety = Yes | Depression = Yes)

(d) Using the Naïve Bayes formula, compute the probability that the new patient has depression.

(e) Predict whether the new patient has depression or not based on the calculated probabilities. **[5 M]**

---------------- ALL THE BEST ---------------